

V3 Discovery Report

Generated: 2026-05-18
Inspector: Abacus AI Agent
Scope: Locate and analyze the “v3” cryogenic dashboard repository

1 — Repository Location

Property	Value
GitHub URL	GBOGEB/document-organization-system (https://github.com/GBOGEB/document-organization-system)
Branch	main
v3 Subfolder	cryo_dashboard_v0_3_0/ cryo_dashboard_v0_3_0/
GitHub Pages	Live Dashboard (https://gbogeb.github.io/document-organization-system/cryo_dashboard_v0_3_0/cryo_dashboard_v0_3_0/)
Status	 ACTIVE — not archived, actively developed
Size	594 KB (GitHub-reported)
Latest Commit	6f08258 — Merge PR #23 (NIST parity tests + deployment checklist)
Latest PR	PR #23 — feature/nist-parity-tests-and-deployment-checklist

2 — What Is v3?

Official Name: Cryogenic Material Property Dashboard
Version: v0.4.9
Organization: SCK CEN
Purpose: Engineering tool for evaluating cryogenic material-property behaviour (1–300 K)

v3 is NOT the same as v4

Aspect	v3 (document-organization-system)	v4 (cryo_leak_rate_dashboard)
Focus	Material properties (k, cp, thermal contraction)	Helium leak rate analysis
Materials	10 materials (AISI316, Al6061, G-10, Cu RRR variants, Ti-6Al-4V)	Helium gas (ideal gas law)
Engine	JavaScript (browser-side computation)	Python (build-time computation)
Runtime	Live interactive dashboard (Plotly.js)	Static HTML + embedded Plotly
Data Source	NIST Monograph 177 coefficients	Contract specs, valve catalogs
Integration	Trapezoid, Simpson, Romberg, Gauss-Legendre	Ideal gas conversions
Panels	12 interactive panels	5 triage pages + 84 HTML files
Version	v0.4.9	v4.0.0

They are complementary — v3 provides material property data that v4's cryogenic system design depends on.

3 — v3 File Structure

```

document-organization-system/
├── .github/workflows/           # CI/CD
├── .nojekyll                   # GitHub Pages config
├── Automation/
│   ├── Scripts/
│   └── Metadata_diffs_log_reviewable.html
├── cryo_dashboard_v0_3_0/
│   └── cryo_dashboard_v0_3_0/  ← THE v3 DASHBOARD
│       ├── index.html          # Landing page
│       ├── dashboard_modular.html # Primary runtime (modular)
│       ├── material_properties_dashboard_v1_10.html # Legacy fallback
│       ├── files.html          # File navigator
│       ├── html_preview_hub.html # Preview hub
│       ├── ssot_launcher.html  # SSOT launcher
│       ├── index_slides.html   # Presentation slides
│       ├── style.css           # Styling
│       ├── js/
│       │   ├── app.js          # Legacy app
│       │   ├── app_modular.js  # Main orchestration (84 KB)
│       │   ├── materials.js    # Material database + evaluators
│       │   ├── numerics.js     # Integration methods
│       │   ├── plots.js        # Plotly rendering
│       │   ├── export.js       # CSV/JSON/PNG export
│       │   ├── state.js        # State management
│       │   ├── logger.js       # Logging
│       │   └── debug.js        # Debug panel
│       ├── data/
│       │   └── materials.json   # Material property database
│       ├── schemas/
│       │   └── materials.schema.json # JSON Schema
│       ├── tests/
│       │   ├── numerics.test.js
│       │   ├── export.test.js
│       │   ├── materials.validate.js
│       │   ├── file_index_integrity.test.js
│       │   ├── static_entrypoints.test.js
│       │   └── nist_parity.test.js # 823 assertions
│       ├── docs/
│       │   ├── CHANGELOG.md
│       │   ├── ENGINEERING_HANDOVER.md
│       │   ├── GITHUB_PAGES_DEPLOYMENT_CHECKLIST.md
│       │   ├── NIST_PARITY_TEST_REPORT_v0.4.9.md
│       │   └── rtm/FEATURE_TEST_FILE_RTM_v0.4.9.md
│       ├── ssot.json           # Single Source of Truth
│       ├── file_index.json     # File index
│       ├── file_index.yaml     # File index (YAML)
│       ├── visual_key.yaml     # Visual key
│       ├── VERSION             # "v0.4.9"
│       └── package.json        # npm config
├── docs/analysis/charts/      # Performance analysis charts (7 HTML)
├── performance_optimizations/ # Cache/async/zip optimizations
├── tests/                     # Top-level tests
├── README.md
└── setup_organization.py      # Organization automation

```

4 — v3 HTML Entry Points

Entry Point	File	GitHub Pages URL
Landing	<code>index.html</code>	Open (https://gbogeb.github.io/document-organization-system/cryo_dashboard_v0_3_0/cryo_dashboard_v0_3_0/index.html)
Modular Dashboard	<code>dashboard_modular.html</code>	Open (https://gbogeb.github.io/document-organization-system/cryo_dashboard_v0_3_0/cryo_dashboard_v0_3_0/dashboard_modular.html)
Legacy Dashboard	<code>materi- al_properties_dashboard_v1_10.html</code>	Open (https://gbogeb.github.io/document-organization-system/cryo_dashboard_v0_3_0/cryo_dashboard_v0_3_0/materi-al_properties_dashboard_v1_10.html)
File Browser	<code>files.html</code>	Open (https://gbogeb.github.io/document-organization-system/cryo_dashboard_v0_3_0/cryo_dashboard_v0_3_0/files.html)
Preview Hub	<code>html_preview_hub.html</code>	Open (https://gbogeb.github.io/document-organization-system/cryo_dashboard_v0_3_0/cryo_dashboard_v0_3_0/html_preview_hub.html)
SSOT Launcher	<code>ssot_launcher.html</code>	Open (https://gbogeb.github.io/document-organization-system/cryo_dashboard_v0_3_0/cryo_dashboard_v0_3_0/ssot_launcher.html)
Slides	<code>index_slides.html</code>	Open (https://gbogeb.github.io/document-organization-system/cryo_dashboard_v0_3_0/cryo_dashboard_v0_3_0/index_slides.html)

Entry Point	File	GitHub Pages URL
		organization-system/ cryo_dashboard_v0_3_0/ cryo_dashboard_v0_3_0/in- dex_slides.html)

5 — v3 Version Lineage

Version	Date	Milestone
v0.1.0	2026-03	Tabbed dashboard prototype
v0.2.0	2026-03	14 engineering improvements in single file
v0.3.0	2026-04	Modular production-style rebuild
v0.4.1	2026-05	Integration method comparison panel
v0.4.2	2026-05	Gauss-Legendre 4-point method
v0.4.3	2026-05	Plot PNG export
v0.4.4	2026-05	Result mode toggle
v0.4.5	2026-05	Delta Summary panel + regression tests
v0.4.6	2026-05-04	Thermal contraction, KPI strip, dual-axis, dark/light theme
v0.4.7	2026-05-05	Comparison view modes, adaptive legend, cursor pins
v0.4.8	2026-05-05	Maintenance release
v0.4.9	2026-05-14	SSOT integration, slideshow presentation, launcher hub

6 — v3 Technical Capabilities

Materials Database (10 materials)

- **Steels:** AISI 316
- **Aluminum:** Al 6061-T6
- **Composites:** G-10 CR Normal, G-10 CR Warp
- **Copper:** OFHC Cu RRR 50/100/150/300/500
- **Titanium:** Ti-6Al-4V

Properties

- **k(T):** Thermal conductivity (W/m·K)
- **cp(T):** Specific heat (J/kg·K)
- **Y(T):** Thermal contraction ($\times 10^{-5}$)

Integration Methods

- Composite Trapezoid ($O(h^2)$)
- Simpson's 1/3 Rule ($O(h^4)$)
- Romberg Integration (adaptive)
- Gauss-Legendre 4-point ($O(h^8)$)

Engineering Formulas

- Conduction heat load: $\dot{Q} = (A/L) \times \int k(T) dT$
- Cooldown energy: $E = m \times \int cp(T) dT$
- Thermal strain: $\Delta L = L_{ref} \times [Y(T_2) - Y(T_1)] \times 10^{-5}$
- Layered wall screening: $R_{total} = \sum_i (L_i / (k_i \times A_i))$

Test Suite

- **823 NIST parity assertions** (nist_parity.test.js)
- Numerics regression tests
- Export consistency checks
- Material schema validation
- File index integrity
- Static entry point existence

7 — Key Finding

The v3 dashboard (document-organization-system) is a fully independent, actively maintained project.

It is NOT a “previous version” of v4 — it is a **complementary tool** focusing on material properties rather than leak rates.

Both projects serve the MYRRHA QPLANT cryogenic engineering ecosystem and should remain as separate, cross-linked repositories.